

Local boundary rigidity for monotone scalar transfer

Research note: a partial result, not the general transfer theorem

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For a $K \times m$ real array $x = (x_j(q))$, assume each column is weakly increasing and all distinct-coordinate mixed moments depend only on their degree:

$$K^{-1} \sum_q \prod_{j \in S} x_j(q) = \mu_{|S|}.$$

The scalar-transfer question asks whether there are K real numbers t_q with $K^{-1} \sum_q t_q^s = \mu_s$ for $1 \leq s \leq m$, and, when the array lies in an interval, whether the t_q can be chosen in that interval. The result below concerns regular scalar boundaries only.

Lemma 1. *If f, g are increasing functions on a probability interval, both with mean zero, then*

$$E|fg| \leq 3E(fg).$$

In particular their covariance is nonnegative; if both are nonconstant, it is strictly positive.

Proof. Write an increasing mean-zero function as a positive Stieltjes combination of $h_s(u) = \mathbf{1}_{u \geq s} - (1 - s)$, $0 < s < 1$. For $s \leq t$,

$$E(h_s h_t) = s(1 - t), \quad E|h_s h_t| = s(1 - t)[1 + 2(t - s)] \leq 3s(1 - t).$$

Integrate the first equality and the inequality against the two positive Stieltjes measures, using the triangle inequality for the absolute value. For finite ordered probability spaces this follows by embedding the functions as step functions. The covariance kernel is strictly positive at every pair of interior jumps, proving the last assertion. $\square$

Theorem 1. *Fix $m \geq 2$ and a scalar array $a(q)$ with distinct levels $a_1 < \dots < a_r$, where $r \leq m - 1$. Let their multiplicities be k_l . Suppose a monic polynomial*

$$P(t) = t^m + \sum_{s=0}^{m-1} \gamma_s t^s$$

satisfies $P'(a_l) = 0$ at every level and $P''(a_l) > 0$ at every level whose multiplicity is at least two. Then there is a neighborhood of the scalar array $(a(q), \dots, a(q))$ with the following property. Every monotone exchangeable-moment array in that neighborhood satisfying

$$\mu_s = K^{-1} \sum_q a(q)^s \quad (1 \leq s \leq m - 1)$$

satisfies

$$\mu_m \geq K^{-1} \sum_q a(q)^m.$$

If equality holds, then on each scalar level all but at most one column are constant, and every column has mean a_l on that level. Such an array has exactly the same mixed moments as the scalar array in all degrees at most m .

Proof. All estimates below have constants depending only on the fixed scalar array and m . On the block B_l of k_l rows write

$$x_j = a_l + b_{jl} + e_j, \quad E_l e_j = 0,$$

where E_l is the uniform mean on the block. Every e_j is increasing there. Set

$$C_{lij} = E_l(e_i e_j) \geq 0, \quad S = \sum_l \sum_{i < j} C_{lij}, \quad B = \max_{j,l} |b_{jl}|, \quad \delta = \max_{j,q} |x_j(q) - a(q)|.$$

For any set I of at least two indices, Lemma 1 implies, for any two distinct $i, j \in I$,

$$\left| E_l \prod_{v \in I} e_v \right| \leq 3(2\delta)^{|I|-2} C_{lij}. \quad (1)$$

Thus the difference between any mixed moment of x and the corresponding mixed moment of the block-constant means $a_l + b_{jl}$ is $O(S)$, uniformly for arrays in a fixed small neighborhood.

We claim that $B = O(S)$. Consider the map from the mr variables b_{jl} to all mixed moments of degrees $1, \dots, m-1$ of the block-constant array. Its derivative at zero is injective. Indeed, a vector b in its kernel satisfies, for every $s \leq m-1$ and every s -element set T ,

$$\sum_{j \in T} \sum_l (k_l/K) a_l^{s-1} b_{jl} = 0.$$

Since $1 \leq s < m$, comparing sets differing in one element shows that all the inner sums are equal, and then each is zero. For each fixed j , the first r of these equations form an invertible weighted Vandermonde system, so every $b_{jl} = 0$. The derivative therefore has a bounded left inverse. The prescribed lower mixed moments and (1) now give $B \leq C(S + B^2)$; after shrinking the neighborhood this yields $B = O(S)$.

For any polynomial P , its symmetric multiaffine polarization in m variables evaluated at $a_l + d_1, \dots, a_l + d_m$ is

$$P(a_l) + \sum_{s=1}^m \frac{P^{(s)}(a_l)}{s!} \frac{e_s(d_1, \dots, d_m)}{\binom{m}{s}}. \quad (2)$$

Here e_s is the elementary symmetric polynomial. Average (2) over rows, putting $d_j = b_{jl} + e_j$. The $s = 1$ term vanishes because $P'(a_l) = 0$. The quadratic term is

$$\sum_l \frac{k_l}{K} \frac{P''(a_l)}{m(m-1)} \sum_{i < j} C_{lij} + O(S^2). \quad (3)$$

By (1) and $B = O(S)$, every term of degree at least three has total absolute value $O(\delta S + S^2)$. On a singleton block all C_{lij} vanish. Hence the main term in (3) is at least cS for some fixed $c > 0$, unless there are no repeated levels, in which case $S = 0$. For sufficiently small δ , the full difference is nonnegative, and is strictly positive if $S > 0$.

By exchangeability of mixed moments, the average polarization is $\mu_m + \sum_{s < m} \gamma_s \mu_s$. The lower moments agree with those of a , giving the desired comparison of the degree- m moments. If $S = 0$, then $B = 0$. Strictness in Lemma 1 says that on each block at most one e_j is nonconstant. Every distinct-coordinate product therefore has block mean $a_l^{|T|}$, as asserted. $\square$

Corollary 1 (Few-level rigidity). *Suppose the scalar baseline has r distinct levels and $2r \leq m-1$. Every sufficiently nearby monotone exchangeable-moment array with the same first $m-1$ moments has exactly the same mixed moments in all degrees at most m . On each baseline level at most one column varies, and all column means there equal the baseline level.*

Proof. Use the proof of Theorem 1 with $Q(t) = \prod_{l=1}^r (t - a_l)^2$ in place of P . Its degree is at most $m - 1$, so the expectation of its polarization is fixed by the prescribed lower moments. Also $Q'(a_l) = 0$ and $Q''(a_l) = 2 \prod_{v \neq l} (a_l - a_v)^2 > 0$ at every level. The same expansion gives a difference at least cS for small enough neighborhoods. Since the difference is zero, $S = 0$, and the equality argument in Theorem 1 applies. $\square$

At a regular scalar constrained local minimum of the m -th moment with the first $m - 1$ moments fixed and $r = m - 1$ distinct interior levels, the Lagrange multiplier polynomial has precisely the form in Theorem 1. Its derivative vanishes at all interior levels. Splitting two equal nodes in opposite directions leaves the lower moments unchanged to first order; the scalar second-order necessary condition gives $P''(a_l) \geq 0$ at every repeated level. At a regular boundary with $r = m - 1$ distinct interior levels, the critical points are simple, so this inequality is strict. Thus the theorem rules out a local outward crossing at every such boundary.

More precisely, when $r = m - 1$, the map from the r levels to the first r scalar moments has an invertible weighted Vandermonde derivative. Nearby lower moments determine a nearby scalar branch with the same multiplicities. Applying Theorem 1 uniformly to that branch shows that nearby monotone tensors lie on its allowed side.

This is *not* a proof of general scalar transfer. It does not classify singular scalar boundary strata, endpoint strata, or global components of the tensor moment image. In particular affine contraction of all columns preserves exchangeability and can move a hypothetical real-line counterexample arbitrarily near a fully collapsed scalar array. The regular-boundary result alone cannot exclude that possibility.